0.2% TINUVIN® 770, Ciba [bis(2,2,6,6-tetramethylpiperidin-4-yl) sebacate];

0.2% CHIMASSORB 944, Ciba [polycondensation product of 4,4'-hexamethylene-bis(amino-2,2,6,6-tetramethylpiperidine) and 2,4-dichloro-6-tert-octylamino-s-triazine].

The two test plaques (NOR-1 and NOR-2) each contain 0.05% N,N,-dialkylhydroxylamine;

NOR-1 additionally contains

0.2% of CHIMASSORB® 119; and

0.2% of TINUVIN® 123, Ciba, [bis(1-octyloxy-2,2,6,6-tetramethylpiperidin-4-yl) sebacate].

NOR-2 additionally contains

0.2% of CHIMASSORB® 119; and

0.2% of the compound of Example 73.

The results of the UV testing are given in the table below.

	DE*		Gloss Value		% Gloss Value	
Sample	0 Kj/m^2	3000 Kj/m^2	0 Kj/m^2	3000 Kj/m^2	0 Kj/m^2	3000 Kj/m^2
control	0.0	4.7	66.6	5.4	100	8.1
NOR-1	0.0	4.0	65.5	16.9	100	25.8
NOR-2	0.0	3.8	64.9	45.3	100	69.8

The compound of Example 73 present in test plaques NOR-2 specifically shows greatly improved gloss retention compared to the less effective control system and in fact is also more effective that a related hindered amine compound (TINUVIN® 123) present in test plaques NOR-1. Resistance to color change upon UV exposure is also enhanced.

Polymer blends containing an unsaturated ternary component, such as EPDM blends, are especially benefited with the more efficient instant light stabilizer systems described above.

In all cases, the light stabilized formulations show much greated resistance to photodegradation than unstabilized specimens which fail quickly under the UV exposure conditions outlined above.

EXAMPLE 77

Paintable TPO

Molded test specimens are prepared by injection molding thermoplastic olefin (TPO) pellets containing the instant compounds, pigments and other coadditives as described in Example 76.

The light stable formulations are painted with one-pack paint systems and tested for TPO/paint interactions. Before painting, the test specimens are first washed in accordance with GM998-4801 and dried for 15 minutes at 200° F. (94° C.). Adhesion promoter is applied to the dry film thickness of 0.2–0.4 mils. The samples are dried for five minutes before a 1K basecoat is applied to a film thickness of 1.2–1.4 mils. The painted panels are dried for three minutes, a clearcoat is then applied to a dry film thickness of 1.2–1.5 mils followed by ten minutes flash drying and a 30 minute oven bake at 250° F. (121° C.).

Paint adhesion is measured by Aggressive Adhesion Testing (proprietary test procedure conducted at Technical Finishing, Inc.) and Taber Scuff. Painted panels which retain greater than 80% of the paint finish are considered acceptable. After Aggressive Adhesion Testing, samples with less than 5% paint loss are deemed acceptable.

Samples are tested to evaluate the TPO/paint interactions as follows:

Formulation*	Taber Scuff Test	Aggresive Adhesion Test	HALS pK_a
A	100% removed	6% Loss (fail)	9.1
B	0% removed	4% Loss (pass)	4.6
C	0% removed	3% Loss (pass)	4.0

Formulation A contains 0.2% CHIMASSORB® 944, 0.2% TINUVIN® 328, 500 ppm calcium stearate and 750 ppm N,N-dialkylhydroxylamine in reactor-grade TPO.

A also contains 0.2% of bis(2,2,6,6-tetramethylpiperidin-4-yl) sebacate (TINUVIN® 770, pK_a of 9.1).

Formulations B and C contain 0.2% CHIMASSORB® 119, 0.2% TINUVIN® 328, 500 ppm calcium stearate and 750 ppm N,N-dialkylhydroxylamine in reactor-grade TPO. **1**

B also contains 0.2% of bis(1-octyloxy-2,2,6,6-tetramethylpiperidin-4-yl) sebacate (TINUVIN® 123, pK_a of 4.6).

C also contains 0.2% of 1-(2-hydroxy-2-methylpropoxy)-4-octadecanoyloxy-2,2,6,6-tetramethylpiperidine (compound of Example 73, pK_a of 4.0)

The data in the table indicate that, although formulation A failed in both the Taber Scuff and Aggressive Adhesion Tests, both formulations B and C passed both paint adhesion tests. However, as inspection of the pK_a values attests, the lower the pK_a value (less basic) for the test hindered amine compound the less paint loss results in this Aggressive Adhesion Test. The instant compound of Example 73 having the hydroxyl moiety present has the lowest pK_a value and also the least paint loss even better than the close prior art compound where no hydroxyl moiety is present.

EXAMPLE 78

Stabilization of Polypropylene Molded Articles

Molded test specimens are prepared by injection molding polypropylene pellets containing pigments, a phosphite, a phenolic antioxidant or hydroxylamine, a metal stearate, ultraviolet light absorbers or a hindered amine stabilizers or a mixture of UV absorbers and hindered amine stabilizers.

Pigmented polypropylene pellets are prepared from pure pigment or pigment concentrates, stabilizers, co-additives and commercially available polypropylene by mixing the components in a Superior/MPM 1" single screw extruder with a general all-purpose screw (24:1 L/D) at 475° F. (250° C.), cooled in a water bath and pelletized. The resulting pellets are molded into 60 mil (0.06 inch thick) 2"×2" plaques at about 475° F. (250° C.) on a BOY 30M Injection Molding Machine.

Pigmented polypropylene formulations composed of polypropylene homopolymer or polypropylene copolymer are stabilized with a base stabilization system consisting of an N,N-dialkylhydroxylamine or a hindered phenolic antioxidant with or without an organophosphorous compound.

All additive and pigment concentrations in the final formulations are expressed as weight percent based on the resin.